

59Antennas.com

DIY kit motor antenna

SWR meter & Motor controller



- SWR meter for Experimental DIY kit
- Motor controller power supply for DIY kit 59Antennas
- Voltage 13.8V input 8V/9.8V/10.5V output switchable
- Color display for voltage and SWR meter
- S1.6 and Best Key for expansion in future
- Copyright 2025 author Volker Hois

DC 13.8 volts must be precise. Antenna output, TX – transceiver



Start screen - bottom right version number firmware v23



SWR meter without connected antenna approx. 10.2 volts



Antenna longer – (lower frequency) approx. 10.5V



Antenna shorter – (higher frequency) approx. 8V



Buttons S1.6 and Best still have functions but are not yet included in v23. Future use.

SWR meter electronic inside
Wiring diagram is on the last 2 pages



Danger!

**Electronic has no fuse, you have to have one from the outside
Integrate backup. The antenna has the inner conductor as a plus!**

You still have to plug in the rear connector (jumper).

On the left side the LM324 is active for 12 volt internal voltage.

The right pin is directly after the 12 volt regulator which can happen

I can't say if you feed in the voltage of 12V here

but it is executed. The 6-pin header is the programming connector

If anyone wants to rewrite the firmware they can do so.

SWR-High with SWR 3.87 in red color

The bar changes color depending on the SWR ratio.

The motor antenna cannot be tuned in the interior

A $\frac{1}{4}$ wave emitter and requires appropriate counterweight.

Tune with low power 5 watts is enough.

Forward 0.52V shows the measured forward voltage.

Backward 0.30V shows the measured reverse voltage.

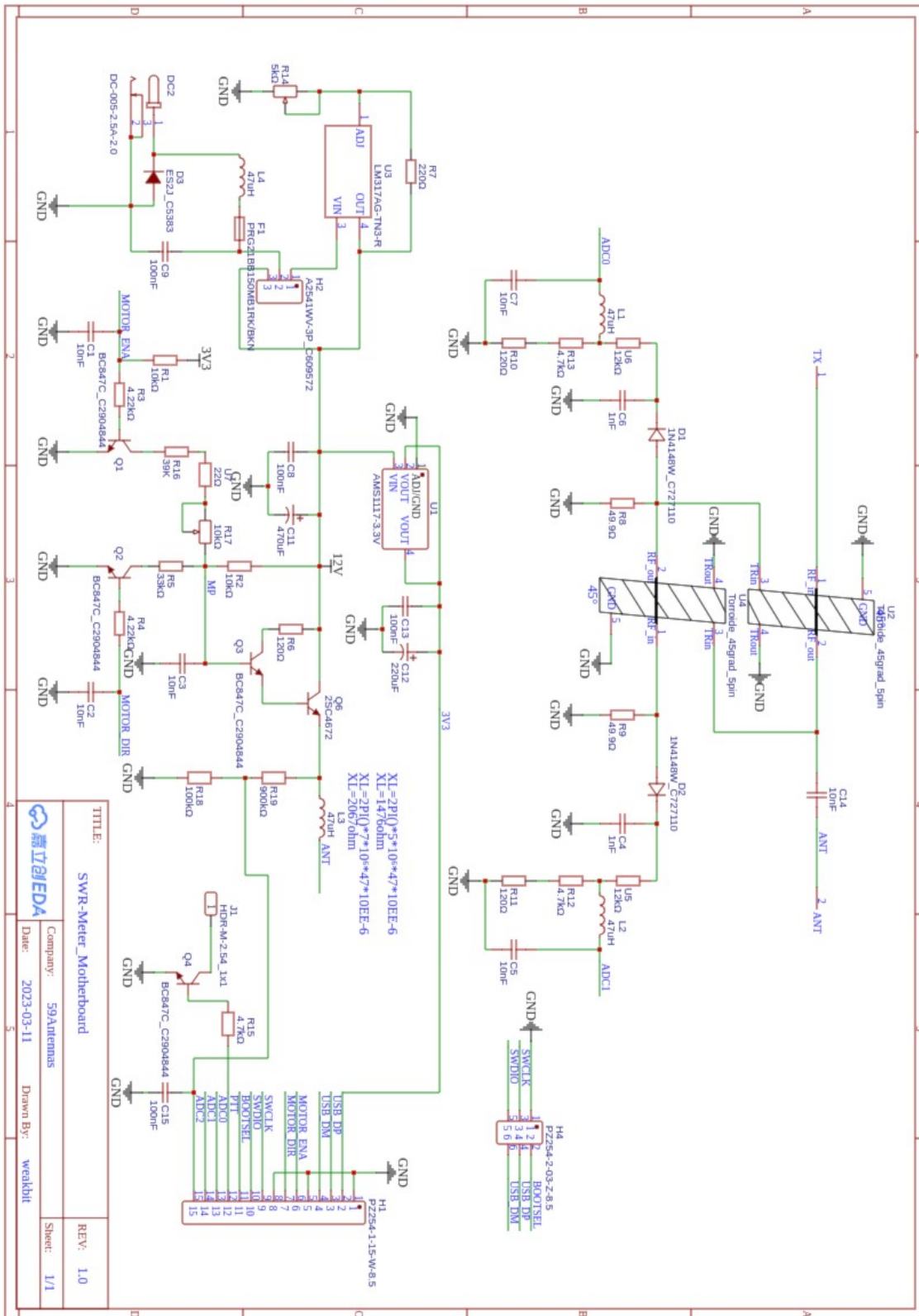
Ant.Out shows the control voltage with 9.6V is stop (approx. 9V – 9.9V)



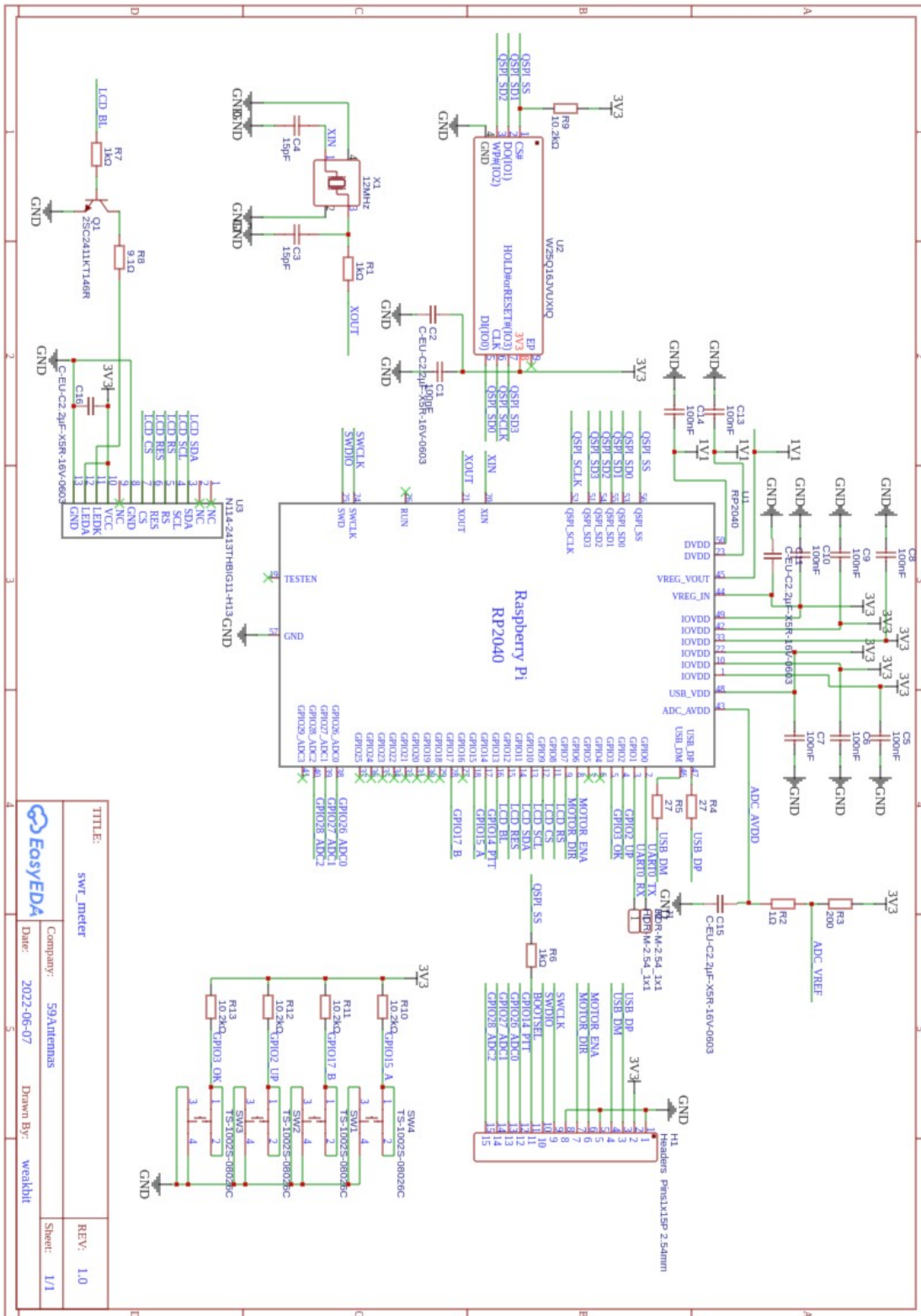
SWR-Low with SWR1.03 in green color



SWR meter motherboard measuring mechanism motor controller



SWR meter micro controller board control



Since it is an experimental kit you can use the electronics like this
rebuild and/or adapt as you see fit. There give
I give everyone a free hand. Above all, it is an experiment.

Tip I saw -

if you live in a house where you cannot connect to ground
I saw an amateur who has 3 radials on the base
screwed on and they are positioned 90° away from the antenna.
Weight created. I can't yet figure out how to tune it
Saying that I have been running the experiment for 7 years and we
We measured it once on open ground and there were the results
Amazing for the extremely short antenna when it comes to SWR.

This kit is just an experiment and is far from having anything
the performance like a taut wire or even a beam antenna.
You have to say that because this kit is intended for people
who have no possibility of making shortwave at all.